

Test Plan & Results Document

Prepared by: Shaden Bawazir

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Project: AI-Powered Appointment Management Assistant

1. Executive Summary

The purpose of this document is to outline the testing strategy, test cases, and results for the **AI-Powered Appointment Management Assistant** project. The testing approach aimed to validate the **Agentforce conversational flow**, the **Salesforce Flow logic**, and the **Lightning component functionality**.

Key Findings:

- All core features (Scheduling, Cancellation, Missing Information Handling) were tested successfully.
- The agent correctly understood natural language requests and executed the corresponding Flows.
- Security configurations (Profiles, Permission Sets, OWD) were verified and working as expected.
- The testing achieved a **100% Pass Rate** across all executed test cases.

2. Test Objectives

- Validate the Agentforce agent's ability to understand natural language requests.
- Verify the Salesforce Flows execute correctly (Create, Update, Cancel).
- Ensure the agent requests missing information before executing transactions.
- Verify security settings are properly configured.

3. Test Environment

Platform: Salesforce Developer Edition

Tools Used:

- Agentforce Builder (Preview)
- Flow Builder (Debug)
- Reports & Dashboards

Test Data:

- Sample appointment records (Appointment c)
- Test users with appropriate profiles (System Administrator, Agent User)

4. Test Data

The following sample data was used for testing:

Record	Customer Name	Appointment Date	Service Type	Status
APT-0001	Sara Ahmed	2026-08-21	Consultation	Scheduled
APT-0002	Ahmed Ali	2026-08-22	Support	Scheduled
APT-0003	Joud	2026-08-23	Consultation	Cancelled
APT-0004	Nora Khalid	2026-08-24	Follow-up	Scheduled

5. Test Cases & Results

Test ID	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status
TC-01	Natural Language Scheduling	1. Open Agent Preview. 2. Type: "Book an appointment for Ahmed at 10:00 AM on 2026-08-21 for Consultation. His email is ahmed@email.com."	Agent understands the request, invokes the Schedule Flow, and confirms the booking.	Appointment created successfully and Agent confirmed.	Pass
TC-02	Missing Information Handling	1. Open Agent Preview. 2. Type: "Book an appointment for Ahmed at 10:00 AM for Consultation." (Missing Email and Date).	Agent asks for the missing details (Email and Date) to continue.	Agent requested missing details.	Pass

TC-03	Natural Language Cancellation	1. Open Agent Preview. 2. Type: "Cancel appointment a00bm00002jdLPqAAM".	Agent routes to Cancel Subagent, invokes Cancel Flow, and confirms status change.	Appointment status updated to "Cancelled".	Pass
TC-04	Flow Execution (Debug)	1. Open Schedule Appointment Flow. 2. Enter Debug input values. 3. Run.	Flow creates a new record successfully.	Flow Debug completed successfully (green status).	Pass
TC-05	Security Check	1. Verify Appointment__c is Private in OWD. 2. Verify Agent has Permission Set access.	User data is secure; Agent can execute operations.	Security configured correctly.	Pass

6. Test Result Analysis

6.1 Quantitative Metrics

Metric	Value
Total Test Cases	5
Passed	5
Failed	0
Pass Rate	100%
Average Execution Time	~2 minutes per case

6.2 Summary of Findings

- **100% Pass Rate:** All test cases (TC-01 to TC-05) were executed successfully.
- **Agentforce Intelligence:** The agent correctly classified requests to the appropriate Subagent (e.g., Appointment Scheduling) and executed the corresponding Action.
- **Flow Integration:** The Flows executed without errors when invoked by the Agent, confirming the integration between Agentforce and Flow Builder is functioning correctly.

- **Security Verification:** The permission sets and OWD settings were verified, ensuring the Agent has the necessary access to perform its tasks.

6.3 Trends Observed

- The agent consistently asked for missing information (email, date) before creating appointments, demonstrating robust conversational AI behavior.
- The **GROUNDED** output evaluation in Agentforce confirmed that the agent's responses were based on real data and not hallucinations.

7. Test Automation & Continuous Integration

While manual testing was performed for this project, **automation tools** can be integrated in the future:

Tool	Purpose
Apex Test Classes	Automate unit testing for custom triggers and classes.
Salesforce CLI	Integrate with CI/CD pipelines for automated deployment and testing.
Selenium	Automate UI testing for Lightning Web Components.

8. Recommendations for Future Testing

- **Additional Test Cases:** Add more test cases for edge cases, such as invalid email format, past dates, or overbooking scenarios.
- **User Acceptance Testing (UAT):** Conduct UAT with real users to validate the user experience and gather feedback.
- **Performance Testing:** Test the agent's response time under high-volume scenarios.
- **Regression Testing:** Automate regression tests to ensure new features don't break existing functionality.

9. Conclusion

The **Test Plan & Results Document** demonstrates a comprehensive testing approach for the AI-Powered Appointment Management Assistant. All test cases were executed successfully, achieving a **100% Pass Rate**. The testing confirmed that the Agentforce agent and Salesforce Flows work together seamlessly, providing a reliable and user-friendly solution.